Check for updates

Journal of Field Robotics

WILEY

RESEARCH ARTICLE

Improved Target Detection in Unmanned Aerial Vehicles Search and Rescue via V-Circular Trajectory Optimization

ShuangShuang He[1] | KaiYu Wang[2] | ZhanZhan Shi[1] | HongYing Qin[1] | Guo Huang[1]

[1]School of Electronic Information and Artificial Intelligence, Leshan Normal University, Leshan, China | [2]School of Microelectronics and Communication Engineering, Chongqing University, Chongqing, China

Correspondence: ShuangShuang He (619880398@qq.com) | Guo Huang (79017771@qq.com)

Received: 26 January 2025 | **Revised:** 31 May 2025 | **Accepted:** 14 February 2026

Funding: Sichuan Science and Technology Seed Program, Grant/Award Numbers: MZGC20240148, MZGC20240035; Anhui Science and Technology Program, Grant/Award Number: 2024AH051751; Sichuan Science and Technology Key Program of Research and Development in Artificial Intelligence, Grant/Award Number: 24ZDYF0017; Leshan Normal University Scientific Research Start-up Project for Introducing High-level Talents, Grant/Award Number: RC2025015

Keywords: target detection | trajectory optimization | UAV search and rescue

ABSTRACT

Unmanned aerial vehicles (UAVs) have become indispensable in search and rescue (SAR) operations during emergencies such as natural disasters and maritime accidents, owing to their extended visual range and operational flexibility. However, UAV-captured images of ground targets are often subject to false positives and false negatives due to suboptimal image quality and target occlusion. In target detection models, detection accuracy is primarily reflected by detection confidence. While improving detection algorithms remains a common approach, this paper emphasizes the significant impact of flight trajectory on detection performance. Specifically, it addresses the trade-off between achieving a wide field of view at high altitudes and obtaining high-resolution images at low altitudes. By analyzing image data collected through circular and oblique descending trajectories, we investigate the effects of flight altitude and viewing angles on detection confidence and establish a Gaussian-based mathematical model to characterize its spatial distribution. To mitigate the probabilities of false negatives and false positives during detection, we propose an optimized flight trajectory planning method termed the V-circular. This approach involves the UAV descending obliquely toward a potential target, conducting circular flights at the lowest altitude to capture high-quality images from multiple viewing angles, and then ascending obliquely to rejoin its predefined trajectory. The V-circular method not only enhances image quality and detection confidence but also enhances the stability of recognition performance. Simulation and experiment results demonstrate that, compared to conventional flight trajectories, the V-circular significantly improves target detection accuracy and shortens SAR time. These advantages are particularly pronounced in missions demanding high precision. Consequently, the V-circular presents an innovative solution for enhancing UAV rescue performance in complex environments.

1 | Introduction

1.1 | Research Background

The increasing frequency of global natural disasters and the rising occurrence of emergencies such as maritime accidents have underscored the critical need for rapid and effective search and rescue (SAR) operations. Unmanned aerial vehicles (UAVs) have emerged as indispensable tools in addressing these challenges due to their extended range of vision and operational flexibility (Fan et al. 2020). Equipped with advanced sensors and global navigation satellite systems (GNSS), UAVs facilitate precise attitude control (Ruan et al. 2023), autonomous flight, obstacle avoidance (Micheal et al. 2021), and ground target recognition and localization through onboard imaging systems (Masadeh et al. 2022; Minaeian et al. 2018).

© 2026 Wiley Periodicals LLC.

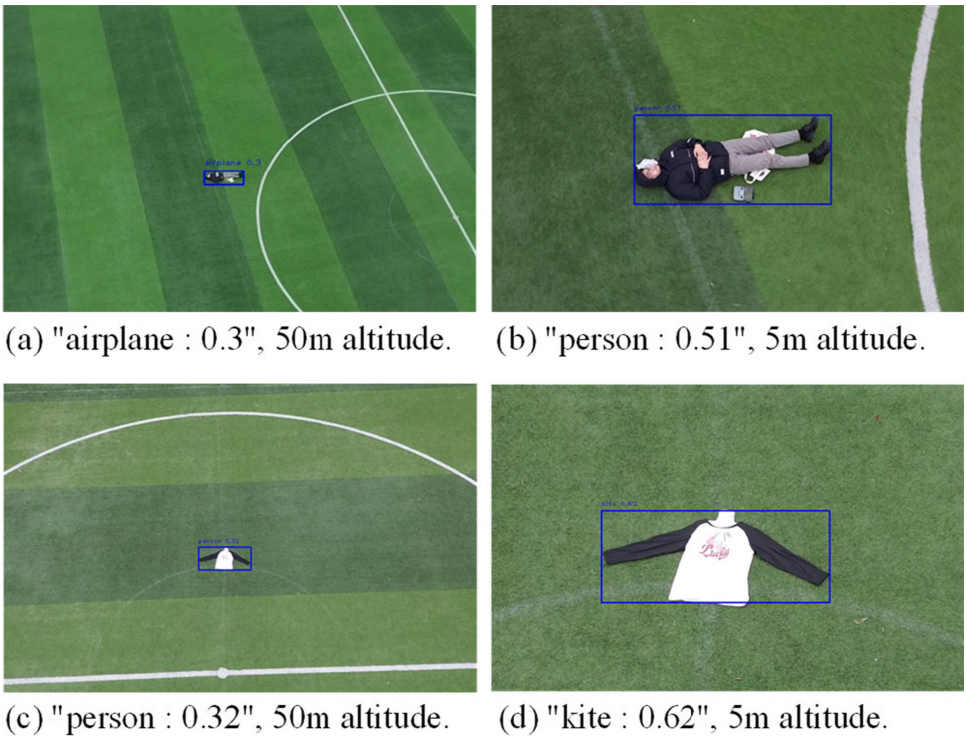


FIGURE 1 | The impact of image quality on detection confidence. (a) "airplane: 0.3," 50 m altitude. (b) "person: 0.51," 5 m altitude. (c) "person: 0.32," 50 m altitude. (d) "kite: 0.62," 5 m altitude. [Color figure can be viewed at wileyonlinelibrary.com]

While most existing research focuses on optimizing target detection algorithms, the impact of UAV flight trajectories on detection performance has received comparatively less attention. Figure 1 illustrates detection results for the same target at different altitudes using the YOLOv11m model in an open field. Figure 1 highlights that low-quality images may result in false positives and false negatives during detection. Higher flight altitudes offer safer operation conditions, broader fields of view, and shorter predefined trajectories, but often compromise image quality, detection confidence, and increase false positive rates (Li et al. 2020). In practical applications, UAV missions often encounter complex terrain and environmental factors, such as forests, grasslands, and seas, which may obscure target and complicate detection. Flight altitude and angle adjustments can help avoid occlusion, improve target visibility, and enhance detection performance without heavy reliance on image processing algorithms.

1.2 | Research Questions

To address the issues of low image quality and frequent target occlusion in UAV-based image acquisition, this study focuses on the impact of flight trajectory on the accuracy of target detection. Image data were collected under circular and oblique descending flight trajectories, and analyzed using the latest YOLOv11m model. Based on these results, we developed a mathematical function that models the distribution of detection confidence across different altitudes and viewing angles. This model enables simulation of detection performance under arbitrary flight trajectories.

Previous studies by our team have shown that V-shaped trajectories can effectively reduce mission time while maintaining information completeness (Li et al. 2022), and that trapezoidal oblique descending trajectories are energy-efficient for UAV trajectory planning (Li et al. 2023). Inspired by these findings and guided by the characteristics of the confidence distribution, this paper proposes a novel trajectory scheme named V-circular, which integrates V-shaped and circular strategies. The workflow is as follows: upon initial detection of a potential target, the UAV performs oblique descending flight approach for preliminary identification; if the target cannot be confidently confirmed, it conducts multi-angle detection around the target at the lowest altitude; finally, the UAV returns to its original height along an oblique ascending trajectory to continue surveillance. This strategy enables close-range and multiview target detection, significantly improving both accuracy and stability.

To evaluate the performance of V-circular, we conducted validation experiments based on both the proposed fitted confidence model and real-world experiments. The results show that the confidence model aligns well with actual observations. Compared with existing detection trajectories, V-circular not only maintains high detection accuracy but also significantly reduces detection time under various confidence thresholds, demonstrating strong practical potential.

This paper makes three major contributions:

- Development of a UAV Target Detection Confidence Distribution Model: By analyzing detection confidence under varying altitudes and viewing angles, this study derives a mathematical function that characterizes confidence variation, providing a theoretical basis for trajectory optimization.
- Proposal of the V-circular Trajectory Strategy: A novel trajectory integrating V-shaped and circular flight patterns

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

is introduced, enhancing detection accuracy while balancing task efficiency.

- Comprehensive Experimental Validation: Through simulations and real-world experiments, the effectiveness and superiority of V-circular are verified in terms of trajectory design, detection accuracy, and detection time, highlighting its applicability in complex environments.

The remainder of this paper is organized as follows. Section 2 introduces the system model and problem formulation, detailing the mathematical model of the detection confidence distribution under two kinds of UAV trajectories. This paper presents the proposed V-circular flight trajectory for target detection in Section 3. Section 4 describes the experiments and analyses conducted to evaluate the proposed scheme. Finally, Section 5 concludes the paper with a summary of findings.

1.3 | Related Work

Current research predominantly focuses on improving image quality and optimizing target detection algorithms. Model optimization has been extensively explored. Under conditions of insufficient lighting or occluded views, Zhao et al. (2021) combined visible-light and infrared images using a generalized convolution operator to preserve information integrity during feature extraction. Chen et al. (2023) reviewed the research status of UAV-based YOLO technologies (YBUT), highlighting their applications in engineering, transportation, agriculture, and automation. Key improvements to YBUT include enhancing detection accuracy and speed and adapting algorithms for diverse application scenarios, such as small target detection and object identification in complex environments. Zheng et al. (2025) proposed the LAM-YOLO model to address the challenges of detecting small target in UAV imagery. By incorporating illumination-occlusion attention mechanisms, their model significantly improved small object recognition. He et al. (2023) tackled occlusion and complex backgrounds in wild facial recognition by employing information reconstruction and occlusion dictionary learning, resulting in significantly improved detection accuracy. Furthermore, Wu et al. (2025) combined spatiotemporal feature fusion with an adaptive viewpoint compensation approach for multiview multi-object tracking in low-altitude scenarios, achieving a detection accuracy of 89.5% in aerial multi-object tracking tasks.

In the case of poor image quality, Valanarasu et al. (2022) developed the TransWeather model, which uses a Transformer-based architecture to mitigate weather-induced image degradations and enhance computer vision performance. Jin et al. (2021) combined super-resolution reconstruction with an improved Faster R-CNN network in their SRD-Net, improving the accuracy of pedestrian detection in low-quality surveillance videos. Additionally, Fu et al. (2024) introduced a dual-learner strategy to tackle image quality degradation under harsh conditions, achieving superior performance across multiple datasets.

Further improving the accuracy of object detection models has become increasingly challenging, researchers have turned to alternative strategies that enhance image acquisition quality to indirectly boost detection performance. Li et al. (2020) experimentally demonstrated that poor image quality leads to false positives, whereas clearer image features significantly improve detection accuracy. Guo et al. (2025) designed a camera system capable of vertical extension and 360° horizontal rotation for wheat disease and pest detection, enabling high-quality image capture from all directions at close range, achieving a detection accuracy of 90%. Liu et al. (2024) proposed a two-stage search strategy based on deep reinforcement learning under a multi-UAV collaboration framework: a high-altitude UAV performs wide-area scanning, while a low-altitude UAV conducts detailed inspections. This method improved target acquisition efficiency in 3D scenarios by 2.19%. Zheng and Ma (2021) defined object detection as a task requiring a specific viewpoint for accurate identification. They set the detection range as the largest cube fully covering the target from that viewpoint, with the center of the cube designated as the detection point.

To reduce image acquisition time, Huang et al. (2024) adopted a spherical sampling strategy for crowd detection in stadiums, where the acquisition area was defined as the surface of a spherical surface centered on the target. Complete information could be obtained once the UAV reached any point on the spherical surface. To expedite the localization of rescue targets, Larsen et al. (2025) considered the search scenario as an unknown environment and applied the A* algorithm for rapid trajectory planning and target localization. Messmer and Zell (2024) integrated real-time meteorological data with probabilistic models, offering a more adaptive and effective solution compared to existing methods. However, despite their efficiency in locating target, both approaches rely on horizontal direct trajectories for each target, resulting in degraded image quality at longer distances.

Current SAR trajectory planning still faces several challenges: some targets can only be effectively detected from specific viewpoints, making it difficult to improve detection performance solely through optimized detection models. While existing trajectory planning approaches capable of rapidly locating target enhance search efficiency, they do not improve image quality. Conversely, strategies aimed at enhancing image quality often fail to meet the minimum detection time requirement.

2 | System Modeling and Problem Presentation

2.1 | System Model

The SAR scene is defined as an open area, which can be forests, grasslands, ocean surfaces, and other terrain. The overall search region is modeled as a square area of size M km $\times$ M km, representing the UAV's operational boundary. Within this region, N potential targets are randomly distributed, including n rescue targets, with the remaining being distractors (e.g., animals, logs, or floating debris). Potential target locations are randomly generated, and their visual features are similar, which increases the complexity of detection and identification.

This paper aims to optimize UAV flight trajectories to improve rescue target detection accuracy by strategically adjusting flight trajectory, altitudes, and viewing angles. As illustrated in Figure 2, the UAV is equipped with a high-precision camera

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

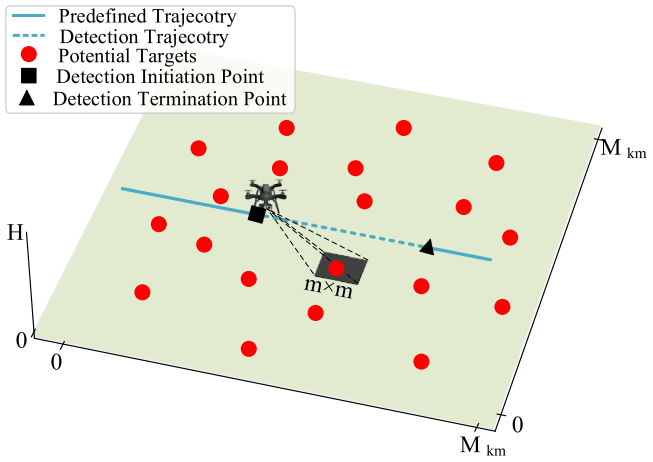


FIGURE 2 | UAV search and rescue system model. [Color figure can be viewed at wileyonlinelibrary.com]

capable of capturing real-time ground images. The UAV operates under constraints including a maximum battery capacity E, a flight altitude H, and a flight speed f_v. The onboard camera can capture ground images covering an $m \times m$ area at an angle directly forward and downward from its position. The dashed line in the figure represents the detection trajectory taken after detecting a target and before returning to the predefined trajectory. It starts at the detection initiation point (DIP) and ends at the detection termination point (DTP), with a straight-line distance of L between them.

In the detection phase, the UAV's high maneuverability allows it to hover, ascend or descend vertically, follow circular trajectories, and perform oblique ascents or descents. These capabilities enable the UAV to dynamically adjust its posture and flight trajectory based on environmental and mission-specific requirements, thereby improving target visibility and detection accuracy from various angles. During SAR missions, the UAV typically follows a snake-like scanning path to ensure complete coverage of the search area.

2.2 | Detection Confidence Modeling

UAVs detect ground targets from the air, with detection confidence influenced by the camera's viewing angle and its relative position to the target. Consequently, the UAV's flight trajectory plays a critical role in determining detection performance. This paper designs two distinct flight trajectories to evaluate target detection:

- Circular trajectory: This trajectory simulates detection tasks performed at a constant altitude while varying the viewing angles around the target.
- Oblique descending trajectory: This trajectory simulates detection tasks conducted at a constant viewing angle while varying the altitudes relative to the target.

The target detection model employs a pretrained YOLO v11m algorithm. The UAV operates at a flight altitude of 50m with a camera viewing angle fixed at 45°. Appendix A illustrates the detection confidence for a potential target using circular trajectories centered on the target. The trajectories are executed with flight altitudes of 50 m and 5 m, respectively. The experimental results indicate that the detection confidence distribution for UAV circular flights closely resembles a Gaussian distribution, characterized by a mean value μ and variance σ_1. This relationship is described mathematically as follows:

$$\text{fn} = X \sim N(\mu, \sigma_1^2), \tag{1}$$

where $X \sim N(\cdot)$ denotes the Gaussian distribution. To ensure that detection confidence remains within the range of (0%, 100%), a sigmoid function is applied as an outer function to the Gaussian distribution. This adjustment normalizes the modeled detection confidence, making it interpretable within the expected range. The confidence distribution model P_R for circular flight can therefore be expressed as:

$$P_R(x, y, z) = \frac{1}{1 + e^{-\alpha\ \text{fn}}}, \tag{2}$$

where α is the gain parameter that controls the rate of change of the sigmoid function, and (x, y, z) represents the relative position of the UAV to the potential target on the ground.

For the oblique descending flight trajectory, the data presented in Appendix B indicates that, although the detection confidence fluctuates locally, it generally increases as the distance to the potential target decreases, eventually stabilizing at a certain level. Using the same curve fitting approach, the target detection confidence model for the oblique descending flight trajectory can be divided into two components:

1. Generation of a set of monotonically nondecreasing random numbers, expressed as follows:

$$h_n = h_{n-1} + |X \sim N(0, \sigma_2^2)|, \tag{3}$$

   where $|X \sim N(0, \sigma_2^2)|$ is the positive increment, indicating that the current value h_n is obtained by increasing the previous value, and σ_2 is the variance.

2. For each time slot, the score follows a Gaussian distribution with h_n as the mean and σ_3 as the variance. The confidence for this point is obtained through Equation (2).

Additionally, the inherent accuracy of the target detection model may introduce some error in target recognition, denoted as ρ. Therefore, the probability that the UAV identifies a potential target on the ground as the actual rescue target at a given location can be rewritten as:

$$P_T(x, y, z) = P_R(x, y, z)\rho. \tag{4}$$

The fitted confidence distributions result, and corresponding parameters for circular flights and oblique descending flights are presented in Appendices A and B, respectively.

In the SAR process, any potential target could be the actual rescue target being sought. Therefore, during the UAV's flight, each potential target should be verified. This study assumes the absence of fraudulent behavior, meaning that there are no false targets posing as the rescue target. To reduce interference from

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

false positives, the final detection accuracy for each potential target must satisfy:

$$P(x, y, z) \geq \varepsilon, \tag{5}$$

here, ε represents the threshold value. When the detection accuracy of a target falls below ε, further verification is required. When detected as other targets, the confidence is expressed by its corresponding negative number.

2.3 | Problem Presentation

In a SAR mission, the objective is to minimize the total SAR time for the area while ensuring that the detection confidence of each target meets the threshold. The UAV rescue process consists of three phases: (1) potential target search time; (2) target verification time when a potential target is identified; (3) UAV battery replacement time, as illustrated in Figure 2. During the target search phase, the UAV follows a predetermined trajectory, such as a snake-like scanning path, performing target detection on the captured ground images. In the second phase, upon detecting a potential target, the UAV adjusts its trajectory to further verify whether it is the rescue target. In the third phase, the UAV performs a battery replacement before complete power depletion. The total SAR time for the UAV can therefore be calculated using the following formula:

$$\min_{(x,y,z)} T = \sum_{j=1}^{S} (T_{f,j} + T_e) + \sum_{i=1}^{N} T_{c,i}, \tag{6}$$

$$s.\ t \quad P(x, y, z) \geq \varepsilon,$$

$$T_{f,j} E_f + \sum T_{c,i} E_c < E,$$

where S represents the number of U-shaped paths in the snake-like scanning trajectory, N is the number of potential targets, $T_{f,j}$ is the time the UAV spends performing a normal patrol along the j-th U-shaped path, T_e is the time required for the UAV to replace its battery, $T_{c,i}$ is the target detection time for potential target i, E_f is the energy consumed by the UAV during normal patrol, E_c is the energy consumed during target detection, and E is the total energy available in the UAV's battery. In the above formula, $\sum_{j=1}^{S} T_{f,j}$ represents the flight time of the UAV along the fixed snake-like trajectory. If full coverage of each area is required, this component cannot be optimized. Similarly, since the location of the battery replacement station is fixed and to ensure the safety of the UAV, it is set that the UAV performs battery replacement after each round trip, $\sum_{j=1}^{S} T_e$ is also nonoptimizable. Therefore, to reduce the SAR time, the only phase that can be optimized is the UAV's target detection time.

In summary, the goal is to optimize the UAV's detection trajectory to minimize the overall SAR time across the entire area, while ensuring that the conditions of detection accuracy and energy constraints are met.

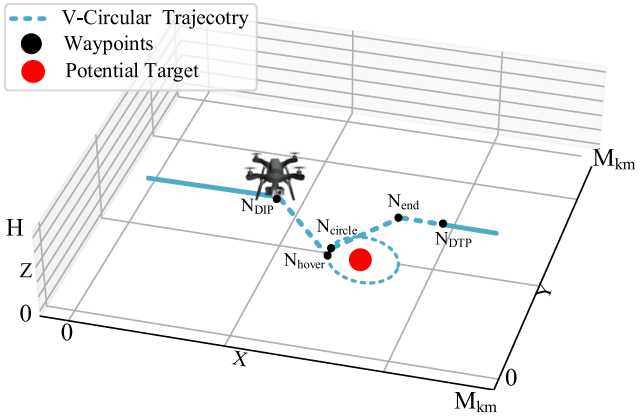


FIGURE 3 | V-circular trajectory of UAV search and rescue. [Color figure can be viewed at wileyonlinelibrary.com]

3 | V-Circular Trajectory Optimization Method

As analyzed in the detection confidence modeling, it can be concluded that to achieve higher detection accuracy, the UAV's detection trajectory should be optimized by adjusting its shooting angle or approaching the target more closely to capture higher-quality images. Based on the derivations in Equations (1) and (3), it is evident that when the UAV follows a circular flight trajectory, its detection confidence follows a Gaussian distribution. By selecting a better angle, the target features can be enhanced. During the UAV's oblique descending flight trajectory, although the angle remains constant, the UAV's proximity to the target allows it to capture clearer images. For convolutional neural network models, this increased proximity results in a higher proportion of the target in the image, leading to a generally nondecreasing trend in detection confidence. Previous studies (Li et al. 2022) have already shown that using a V-shaped trajectory when the UAV is communicating with the ground node optimizes channel quality and minimizes task completion time. Inspired by this, this paper introduces a combined detection trajectory integrating the V-shaped trajectory and the circular trajectory referred to as the V-circular trajectory, as illustrated in Figure 3.

As shown in Figure 3, the V-shaped trajectory enables the UAV to approach the potential target in the shortest time, making it the most energy-efficient flight trajectory (Li et al. 2023). By reducing the distance to the target, the UAV can capture more detailed features, thereby enhancing detection accuracy. If the UAV is unable to accurately identify the target after reaching the lowest altitude, it can transition to a circular trajectory around the target, conducting close-range detection from multiple viewing angles.

As illustrated in Figure 3, the trajectory from N_{DIP} to N_{hover} represents the UAV oblique descending toward the target, while the trajectory from N_{hover} to N_{circle} is a horizontal circular flight trajectory, the trajectory from N_{circle} to N_{end} represents the UAV oblique ascending toward the predefined trajectory. From N_{end} to N_{DTP} represents the remaining detection trajectory after the UAV returns to the predefined trajectory ahead of schedule. Thus, the oblique descending flight time $t_{\text{down}} = \frac{|N_{\text{DIP}} - N_{\text{hover}}|}{f_v}$, the circular flight time $t_{\text{circle}} = \frac{\langle N_{\text{hover}} - N_{\text{circle}} \rangle}{f_c}$, and the oblique ascending flight time $t_{up} = \frac{|N_{\text{circle}} - N_{\text{end}}|}{f_v}$, direct flight time $t_{\text{direct}} = \frac{|N_{\text{end}} - N_{\text{DTP}}|}{f_v}$, where f_v is the UAV's oblique descending and oblique ascending

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 | UAV performance parameters.

Flight speed	5 m/s
Flight height	50 m
Detection range at 50 m height	50 × 50 m
Minimum flight height	5 m
V-circular trajectory pitch angle	45°
Circling radius	Flight height
Direct flight energy consumption (Li et al. 2023)	28.6 J/m
Oblique descending flight energy consumption (Li et al. 2023)	32.1 J/m
Oblique ascending flight energy consumption (Li et al. 2023)	37.7 J/m
Circling energy consumption (Ji et al. 2020)	30 J/m
Vertical downward flight energy consumption (Li et al. 2023)	27.58 J/m
Vertical upward flight energy consumption (Li et al. 2023)	30.3 J/m

flight speed, and f_c is the UAV's circular flight speed. $|\bullet|$ represents the distance between two points in 3D coordinates, and $\langle \bullet \rangle$ represents the arc length with radius r. The UAV's detection time for the i-th potential target is thus:

$$T_{c,i} = t_{\text{down}} + t_{\text{circle}} + t_{\text{up}} + t_{\text{direct}}. \tag{8}$$

The energy consumption of the UAV for detecting the i-th potential target is:

$$E_{c,i} = e_{\text{down}} t_{\text{down}} + e_{\text{circle}} t_{\text{circle}} + e_{\text{up}} t_{\text{up}} + e_{\text{direct}} t_{\text{direct}}, \tag{9}$$

where e_{down}, e_{up}, e_{circle}, and e_{direct} represent the energy consumption of the UAV during the oblique descending, oblique ascending, circular, and direct trajectories, respectively.

Once the UAV detects a potential target, it uses image localization technology (Elhadary et al. 2022) to obtain the coordinates of the target and immediately adjusts its flight trajectory to descend towards the target. If the target cannot be confirmed at the lowest altitude, the UAV will perform a circular flight around the target at that altitude for further detection. The overall process is outlined as follows:

Target detection process of V-circular trajectory

1. The UAV target search following a snake-like trajectory.
2. When a potential target (PT) is detected, $P(x, y, z) < \varepsilon$.
3. **while**($|\text{UAV} - \text{PT}| > \text{rand} \quad P(x, y, z) < \varepsilon$)**:**

   While flying toward the potential target in an oblique descent, the UAV continuously performs detection.

   end while
4. **while** ($\langle N_{\text{hover}} - N_{\text{circle}} \rangle < 2\pi r$ and $P(x, y, z) < \varepsilon$)**:**

   The UAV performs a circular flight around the target and continuously performs detection.

   end while
5. **if** ($P(x, y, z) \geq \varepsilon$)**:**

   if (The target is a rescue target)**:**

   Send instructions to the data center.

Target detection process of V-circular trajectory

else:

Fly at an oblique ascending back to the preset trajectory and return to **step 1**.

end if

4 | Simulation Results and Discussion

4.1 | Simulation Experiments

In this section, numerical simulations are performed to assess the performance of the algorithm in an idealized environment. For simplicity, the UAV flight speed is set to 5 m/s, the flight altitude to 50 m, the horizontal distance of the detection trajectory is L = 110 m, and the search area is defined as 1 × 1 km. The specific parameters are listed in Table 1. The UAV conducts target detection along a predefined snake-like flight path, replacing the battery after completing each U-shaped path. The search area contains 20 randomly distributed potential targets, among which the first 10 are designated as rescue targets, and the remaining 10 are considered false targets. Unless otherwise specified, the target detection model accuracy is set to $\rho = 0.95$, and the probability $P_R(x, y, z)$ for all rescue targets is randomly generated within the range of 0.35 to 0.55, and false is 0.25 to 0.38. The threshold for the final detection confidence of the model is set to $\varepsilon = 0.80$. The parameters used for oblique descending detection confidence in the V-circular method are detailed in Appendix B. The confidence distribution of false targets is illustrated in Appendix C, and the variance parameter is selected based on the variance of the mean value corresponding to the V-shaped trajectory. The detection time for each potential target is defined as the duration from N_{DIP} to N_{DTP}. The simulation parameters are specific to videos captured by the DJI Mini 3 Pro and based on the YOLOv11m.

To assess the effectiveness of the proposed approach, this paper selected two comparative schemes for evaluation:

- Circular Flight: After detecting a potential target, the UAV performs a circular trajectory around the target with a 50m radius and a 45° shooting angle for detection. Once the

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 | Detection data of potential targets under different schemes ($\varepsilon = 0.80$).

Target index	Initial confidence	V-circular		Vertical		Circular	
		Final confidence	Detection time/s	Final confidence	Detection time/s	Final confidence	Detection time/s
1	0.35	0.86	30	0.86	40	0.56	85
2	0.36	0.84	30	0.84	40	0.59	85
3	0.53	0.81	26	0.88	40	0.78	85
4	0.55	0.82	28	0.92	40	0.79	85
5	0.36	0.82	30	0.82	40	0.64	85
6	0.50	0.84	29	0.89	40	0.68	85
7	0.39	0.85	30	0.85	40	0.58	85
8	0.53	0.83	29	0.83	40	0.79	85
9	0.45	0.83	31	0.83	40	0.71	85
10	0.43	0.81	29	0.85	40	0.80	61
11	0.38	0.23	36	0.23	46	0.55	85
12	0.35	0.08	36	0.08	46	0.57	85
13	0.29	0.08	36	0.08	46	0.38	85
14	0.30	0.07	36	0.07	46	0.40	85
15	0.33	0.10	36	0.10	46	0.51	85
16	0.28	0.15	36	0.15	46	0.57	85
17	0.31	0.06	36	0.06	46	0.37	85
18	0.35	0.08	36	0.08	46	0.59	85
19	0.35	0.12	36	0.12	46	0.43	85
20	0.31	0.15	36	0.15	46	0.57	85
Total detection time/s		625		860		1676	

target is confirmed, the UAV returns to its predefined trajectory at the same altitude. The variance is $\sigma_1^2 = 0.12$.

- Vertical Flight: After detecting a potential target, the UAV flies directly above it and then descends vertically to approach the target. The UAV then performs a circular trajectory with a 5 m radius for detection. Once the target is confirmed, the UAV ascends vertically and returns to the predefined trajectory. The confidence is derived from the circular trajectory segment of the V-circular trajectory at 5 m.

Table 2 presents the detection real-time data of potential targets, the proposed V-circular trajectory, and two comparative schemes. It can be observed that among the three UAV trajectory planning methods, the V-circular trajectory achieves the highest target detection confidence through its close-range multiview strategy. Moreover, due to its rapid approach to the target, it also yields the shortest detection time. The Vertical trajectory also provides close-range multiview observation, achieving a confidence comparable to that of the V-circular. However, due to its horizontal flight followed by vertical descent, it covers a longer distance and thus takes more time. The circular trajectory operates at a higher altitude. Despite offering multiple viewing angles, its confidence is lower than that of the close-range methods. Additionally, its larger radius leads to both the lowest confidence and the longest detection time. False targets, which start with low confidence, become even less likely as the UAV approaches and clearer features are observed. Due to its long detection range, the circular trajectory has a higher risk of false or missed detections. When the detection threshold is not met, the UAV must complete the entire trajectory, leading to the maximum detection time. Overall, the V-circular trajectory exhibits greater flexibility in handling targets with lower image quality, offering a more efficient solution in such cases.

Figure 4 illustrates the number of target detection completed by the three schemes under varying detection confidence thresholds. The total includes both rescue targets confirmed and false targets rejected. A lower threshold increases the risk of false target misclassification. As the threshold increases, the confirmed number of targets in all schemes decreases. The V-circular and vertical trajectories perform similarly under high confidence conditions, effectively completing the target detection tasks. The circular trajectory scheme, due to its fixed altitude, results in lower image quality for some targets, and with the increase of the threshold, the number of accurately identified targets decreases more significantly. The results clearly show that the V-circular trajectory, by dynamically adjusting both distance and angle, yields superior target detection performance under higher threshold conditions. However, when the threshold exceeds the inherent capability of the detection model, even the V-circular trajectory fails to confirm the targets.

Figure 5 examines the impact of different detection confidence thresholds on the UAV's total detection time, which is the sum of the detection times for all potential targets. The V-circular

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

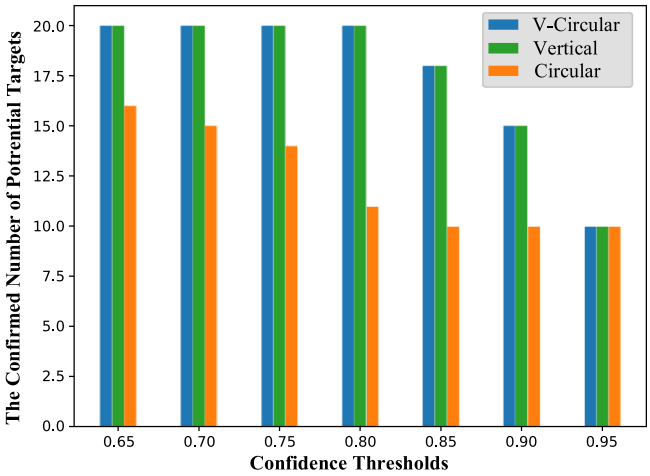


FIGURE 4 | Number of tasks completed at different confidence thresholds. [Color figure can be viewed at wileyonlinelibrary.com]

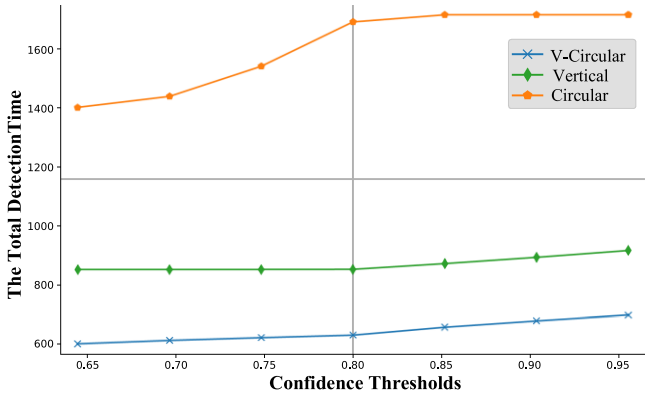


FIGURE 5 | The impact of different confidence thresholds on UAV's detection time. [Color figure can be viewed at wileyonlinelibrary.com]

trajectory, by dynamically adjusting its flight trajectory based on target image clarity, offers high adaptability in trajectory length, thereby maintaining a relatively short detection time while ensuring a high confidence level. The Vertical flight scheme, which involves a vertical approach and circling around the target, results in a longer trajectory path, leading to a detection time that is second only to the V-circular scheme. The circular flight scheme, under high confidence thresholds, requires circling over a longer arc, which significantly increases its detection time, making it the least efficient. In conclusion, the V-circular trajectory provides excellent controllability over detection time while facilitating efficient rescue operations.

4.2 | Real-World Experiments

To further validate the practicality of our approach, real-world experiments are conducted under realistic constraints. A DJI Mini 3 Pro UAV is used with the YOLOv11m object detection model. The ground setup includes 20 potential targets: 10 human subjects in various postures (standing, sitting, and lying) placed across three scenarios—playground, bushy area, and forested region. The remaining 10 are false targets, such as buckets, streetlights, and pieces of clothing, which are also distributed across the same environments, as shown in Figure 6.

Five trajectory planning methods are compared in this real-world setting: circular, vertical, direct (Larsen et al. 2025), spherical (Huang et al. 2024), and square (Zheng and Ma 2021).

- **Circular** evaluates high-altitude circular trajectory for target detection.
- **Vertical** tests low-altitude circular trajectory.
- **Direct** serves as a baseline without additional trajectory optimization.
- **Spherical** verifies confidence levels under minimal detection time, using a spherical trajectory with a 5 m radius.
- **Square** assesses detection confidence from optimal viewpoints, with the detection point located at the center of the cube (i.e., at an altitude of 25 m).

As illustrated in Figure 7, the real-world experiment results show that as the confidence threshold increases, the number of confirmed potential targets decreases across all schemes. At a threshold of 0.95, none of the methods can identify any target, which is consistent with the simulation results. Among the tested methods, vertical and our proposed V-circular trajectory achieve the best performance at all thresholds and yield nearly identical results. This is because, in its later phase, the V-circular trajectory performs low-altitude circumnavigation, the same as the Vertical scheme, both providing close-range multiview images of high quality. At a threshold of 0.65, the Vertical scheme identifies 19 targets, indicating that low thresholds are more susceptible to image noise, leading to the misclassification of one false target as a true positive. The Direct scheme only provides a single, distant viewpoint, resulting in poor image quality and a higher risk of false detections at low thresholds. The circular scheme performs slightly better due to its multi-view capability. The square scheme captures images from specific viewpoints at a lower altitude than the circular, thereby achieving better detection performance. The spherical scheme employs low-altitude single-view detection, yielding improved image quality and better results. In summary, image quality is a key factor affecting detection confidence. Close-range multiview observation significantly enhances image quality and thus improves detection accuracy.

Figure 8 shows the total detection time required for all 20 potential targets under different confidence thresholds in a real-world experiment. It can be seen that the detection time of direct, square, and spherical remain constant because their trajectories are fixed. Direct has no additional trajectory detection and only flies in a straight line at a fixed height, thus taking the shortest time. Square performs detection at an altitude of 25 m and returns to the predefined trajectory via the shortest path, yielding a slightly longer time. Assuming that the Spherical detection trajectory adopts the fastest V-shaped trajectory, but the height of the detection point is 5 m, leading to a moderate increase in time. The proposed V-circular trajectory combines V-shaped and circular flight patterns. At low thresholds, the potential targets can often be confirmed using only the V-shaped trajectory, resulting in a shorter detection time than Spherical. At high thresholds, if the V-shaped fails to confirm the potential targets, the UAV performs a circular flight

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

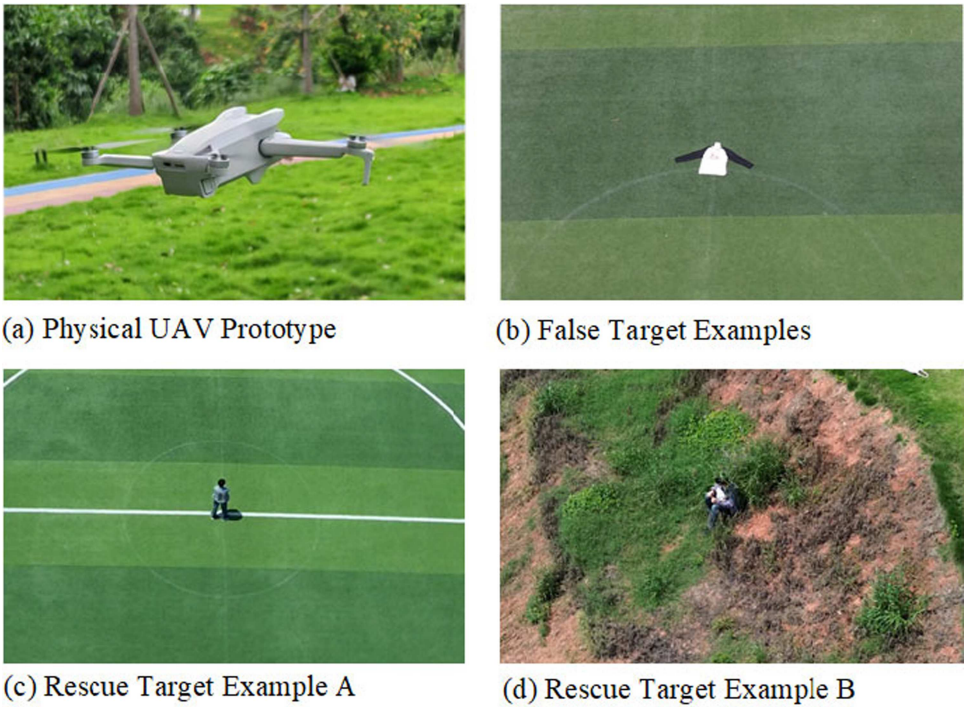


FIGURE 6 | Representative real-world experiment cases. (a) Physical UAV prototype. (b) False target examples. (c) Rescue target example A. (d) Rescue target example B. [Color figure can be viewed at wileyonlinelibrary.com]

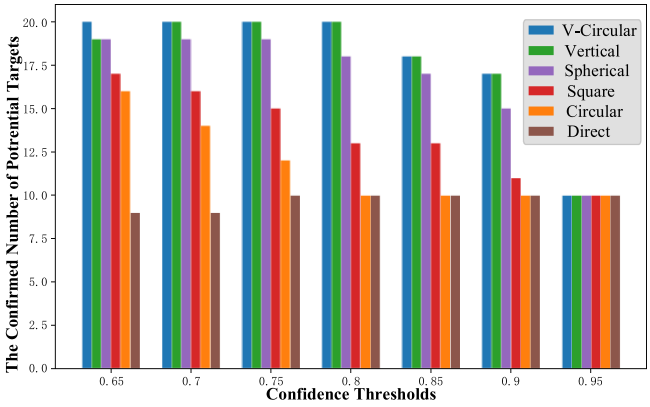


FIGURE 7 | Real-world experiments: Number of tasks completed at different confidence thresholds. [Color figure can be viewed at wileyonlinelibrary.com]

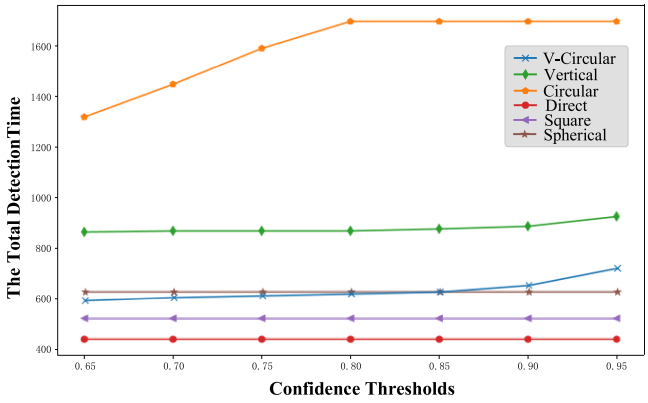


FIGURE 8 | Real-world experiments: The impact of different confidence thresholds on UAV SAR time. [Color figure can be viewed at wileyonlinelibrary.com]

at the lowest point, increasing the total time slightly beyond that of Spherical. The vertical trajectory consists of four fixed trajectories and takes a relatively long time overall. In contrast, circular, due to its longer distance and lower confidence level, usually requires a complete circular trajectory to be able to confirm the potential target, thus taking the longest time.

To sum up, above real-world experiments have verified the correctness of the theory proposed. The results show that improving image quality is conducive to enhancing the detection confidence; Furthermore, even with similar image quality, better viewing angle can also further enhance the confidence. This V-circular scheme combines close-range multi-view angles and dynamic trajectory adjustment, achieving the best balance between detection speed and accuracy based on the existing detection trajectories.

5 | Conclusion

This paper investigates the influence of different altitudes and viewing angles on detection confidence between UAV and ground target, addressing the existing gap in understanding how UAV trajectories affect detection confidence. This work demonstrates that the detection confidence distribution patterns obtained from both circular and oblique descending flight trajectories around rescue targets can be modeled using a Gaussian distribution. The fitting results indicate that close-range, multiview trajectories can enhance detection confidence. Driven by the objective of minimizing rescue time, this paper proposes a V-circular detection trajectory method that dynamically adjusts the UAV's altitude and viewing angle based on the characteristics of the confidence distribution. Compared with existing trajectory methods, simulations and real-world

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

experiments validate the proposed approach's comprehensive advantages in terms of both detection accuracy and SAR efficiency, achieving an effective balance between performance and efficiency.

Future research directions include: (1) developing fast false target rejection mechanisms to reduce overall detection time; and (2) integrating multimodal data, such as thermal imaging and visible-light images, to further improve target detection and trajectory planning strategies.

Acknowledgments

This work was supported in part by the Sichuan Science and Technology Seed Program (Grant No. MZGC20240148), MZGC20240035; supported by Anhui Science and Technology Program (Grant No. 2024AH051751); Sichuan Science and Technology Key Program of Research and Development in Artificial Intelligence (Grant No. 24ZDYF0017); Leshan Normal University Scientific Research Start-up Project for Introducing High-level Talents (Grant No. RC2025015).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

Chen, C., Z. Zheng, T. Xu, et al. 2023. "YOLO-Based UAV Technology: A Review of the Research and Its Applications." *Drones* 7, no. 3: 190.

Elhadary, A., M. Rabah, E. Ghanim, R. Mohie, and A. Taha. 2022. "The Influence of Flight Height and Overlap on UAV Imagery over Featureless Surfaces and Constructing Formulas Predicting the Geometrical Accuracy." *NRIAG Journal of Astronomy and Geophysics* 11, no. 1: 210–223.

Fan, B., Y. Li, R. Zhang, and Q. Fu. 2020. "Review on the Technological Development and Application of UAV Systems." *Chinese Journal of Electronics* 29, no. 2: 199–207.

Fu, C., J. Xiao, W. Yuan, R. Liu, and X. Fan. 2024. "Learning Cruxes to Push for Object Detection in Low-Quality Images." *IEEE Transactions on Circuits and Systems for Video Technology* 34, no. 12: 12233–12243.

Guo, X., Y. Ou, K. Deng, X. Fan, R. Gao, and Z. Zhou. 2025. "A Unmanned Aerial Vehicle-Based Image Information Acquisition Technique for the Middle and Lower Sections of Rice Plants and a Predictive Algorithm Model for Pest and Disease Detection." *Agriculture (London)* 15, no. 7: 790.

He, H., J. Liang, Z. Hou, H. Liu, and X. Zhou. 2023. "Occlusion Recovery Face Recognition Based on Information Reconstruction." *Machine Vision and Applications* 34, no. 5: 74.

Huang, M., H. Li, Y. Zhou, T. Ma, J. Su, and H. Zhou. 2024. "A UAV Aided Lightweight Target Information Collection and Detection Approach." *Peer-to-Peer Networking and Applications* 17, no. 3: 1667–1681.

Ji, X., X. Meng, A. Wang, et al. 2020. "E2PP: An Energy-Efficient Path Planning Method for UAV-Assisted Data Collection." *Security and Communication Networks* 2020, no. 1: 8850505.

Jin, Y., Y. Zhang, Y. Cen, Y. Li, V. Mladenovic, and V. Vornin. 2021. "Pedestrian Detection With Super-Resolution Reconstruction for Low-Quality Image." *Pattern Detection* 115: 107846.

Larsen, A., K. Kilic, M. Ladefoged, and I. Sung. 2025. "A Novel Path-Finding Approach for Maritime Search-and-Rescue Missions Incorporating Dynamic Probability of a Target Location." *Engineering Optimization* 57, no. 11: 3521–3540.

Li, M., S. He, and H. Li. 2022. "Minimizing Mission Completion Time of UAVs by Jointly Optimizing the Flight and Data Collection Trajectory in UAV-Enabled WSNs." *IEEE Internet of Things Journal* 9, no. 15: 13498–13510.

Li, M., H. Li, P. Ma, and H. Wang. 2023. "Energy Maximization for Ground Nodes in UAV-Enabled Wireless Power Transfer Systems." *IEEE Internet of Things Journal* 10, no. 19: 17096–17109.

Li, Q., T. Jussi, Q. Jorge, G. Tuan, G. Moncef, and T. Hannu. 2020. "Towards Active Vision With UAVs in Marine Search and Rescue: Analyzing Human Detection at Variable Altitudes." In *2020 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, 65–70.

Liu, Y., X. Li, J. Wang, F. Wei, and J. Yang. 2024. "Reinforcement-Learning-Based Multi-UAV Cooperative Search for Moving Targets in 3D Scenarios." *Drones* 8, no. 8: 378.

Masadeh, A., M. Alhafnawi, H. Salameh, A. Musa, and Y. Jaraweh. 2022. "Reinforcement Learning-Based Security/Safety UAV System for Intrusion Detection under Dynamic and Uncertain Target Movement." *IEEE Transactions on Engineering Management* 71: 12498–12508.

Messmer, M., and A. Zell. 2024. "Evaluating UAV Path Planning Algorithms for Realistic Maritime Search and Rescue Missions." In *2024 International Conference on Unmanned Aircraft Systems*, 472–479.

Micheal, A., K. Vani, S. Sanjeevi, and C. Lin. 2021. "Object Detection and Tracking With UAV Data Using Deep Learning." *Journal of the Indian Society of Remote Sensing* 49: 463–469.

Minaeian, S., J. Liu, and Y. Son. 2018. "Effective and Efficient Detection of Moving Targets From a UAV's Camera." *IEEE Transactions on Intelligent Transportation Systems* 19, no. 02: 497–506.

Ruan, T., Y. Huang, Q. Zhu, C. Hao, and Q. Wu. 2023. "Multi-Stage RF Emitter Search and Geolocation With UAV: A Cognitive Learning-Based Method." *IEEE Transactions on Vehicular Technology* 72, no. 5: 6349–6362.

Valanarasu, J., R. Yasarla, and V. Patel. 2022. "Transweather: Transformer-Based Restoration of Images Degraded by Adverse Weather Conditions." In *Proceedings of the IEEE Conference on Computer Vision and Pattern Detection*, 2353–2363.

Wu, P., Y. Li, Z. Li, X. Yang, and D. Xue. 2025. "Multi-View, Multi-Target Tracking in Low-Altitude Scenes With UAV Involvement." *Drones* 9, no. 2: 138.

Zhao, F., W. Zhao, L. Yao, and Y. Liu. 2021. "Self-Supervised Feature Adaption for Infrared and Visible Image Fusion." *Information Fusion* 76: 189–203.

Zheng, X., and C. Ma. 2021. "An Intelligent Target Detection Method of UAV Swarms Based on Improved KM Algorithm." *Chinese Journal of Aeronautics* 34, no. 2: 539–553.

Zheng, Y., Y. Jing, J. Zhao, and G. Cui. 2025. "LAM-YOLO: Drones-Based Small Object Detection on Lighting-Occlusion Attention Mechanism YOLO." *Computer Vision and Image Understanding* 261: 100489.

Appendix A

Confidence Fitting for Circular Trajectories

From the confidence distributions at two altitudes shown in Figure A1, it can be observed that detection confidence under varying viewpoints fluctuates around a certain mean value (note: the confidence of non-rescue targets have been effectively filtered out). This is due to significant differences in target appearance across different viewing angles, leading to noticeable fluctuations in the low-confidence range, while the high-confidence range remains relatively stable. Moreover, as shown in Figure A1a,b, the detection confidence of the same target at an altitude of 5 m is significantly higher than that at 50 m, indicating that relative distance is a primary factor influencing detection confidence.

The confidence distribution curves were fitted by counting the number of detections within each confidence interval, as shown in Figure A2a,b. The distributions at both 50m and 5m altitudes exhibit a bell-shaped

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

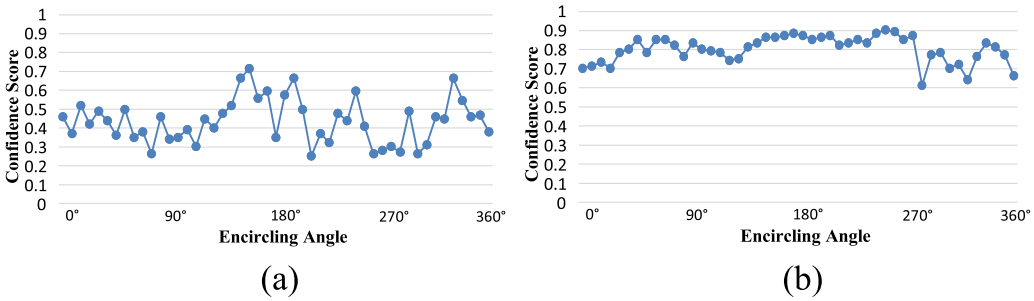


FIGURE A1 | Detection confidence for potential targets using circular trajectories at different altitudes. (a) 50 m altitude. (b) 5 m altitude. [Color figure can be viewed at wileyonlinelibrary.com]

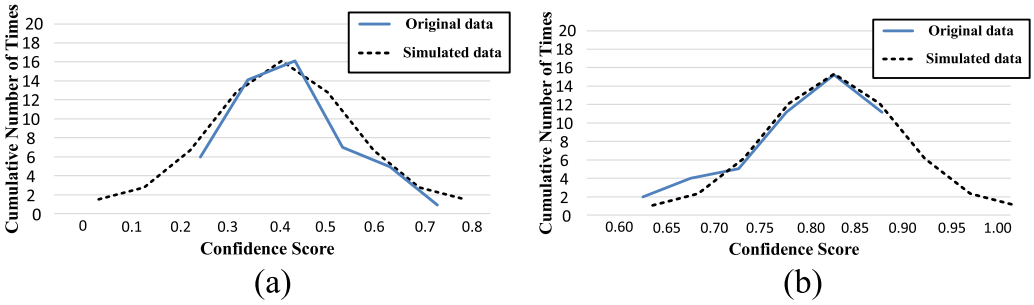


FIGURE A2 | Confidence distribution using circular trajectories at different altitudes. (a) 50 m altitude. (b) 5 m altitude. [Color figure can be viewed at wileyonlinelibrary.com]

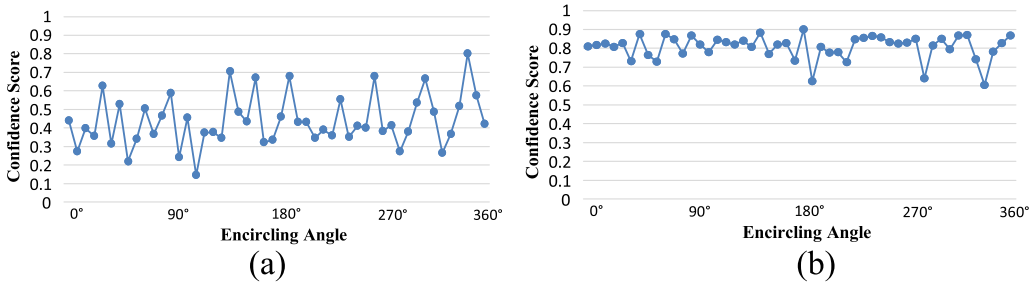


FIGURE A3 | Fitting of confidence distributions using circular trajectories at different altitudes. (a) 50m altitude. (b) 5m altitude.

pattern. The missing left part of the 50 m curve and the right part of the 5 m curve are due to the inherent accuracy limitations of the Yolov11 model. Fitting results show that the detection confidence at 50 m approximately follows a Gaussian distribution with a mean of 0.4 and a variance of 0.02, while at 5 m, it follows a Gaussian distribution with a mean of 0.83 and a variance of 0.06.

The above confidence distributions can be modeled using functions (1), (2), and (4) in the main text, generating the confidence curves for the 50 m and 5 m circular trajectories, as shown in Figure A3a,b. The parameters for 50 m are $\mu = 0.45$, $\sigma_1^2 = 0.12$, $\alpha = -6$; and for 5 m are $\mu = 0.75$, $\sigma_1^2 = 0.09$, $\alpha = -6$. Due to the introduction of the Sigmoid function, the mean value experiences a slight shift.

Appendix B

Confidence Fitting for Oblique Descending Trajectories

For the oblique descending detection trajectory, Figure B1a shows that the detection confidence generally increases as the distance to the potential target decreases, eventually stabilizing at a certain level. This plateau is attributed to the inherent limitations of the YOLOv11 model, which cannot further improve confidence. However, some randomness remains within each altitude interval, likely caused by factors such as UAV camera vibration, lighting conditions, weather, and focusing errors.

The confidence distribution curve for the oblique descending trajectory consists of two steps: (1) generating a set of monotonically non-decreasing baseline confidence values; and (2) generating corresponding confidence values based on each baseline, using it as the mean and σ_3 as the variance. Confidence values fluctuate more around 0.5 and less at both ends, indicating larger variances in the mid-range and smaller variances at both ends.

The simulation uses functions (1), (2), (3), and (4). When the mean is set to $h = 0.5$, σ_3^2 is initialized to 0.1. For every 0.01 increase or decrease in the mean, the variance is multiplied by 0.95. At 50 m altitude, the initial mean is set to $h_1 = 0.2$ with an incremental variance $\sigma_3^2 = 0.09$. The fitted confidence curve is shown in Figure B1b.

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

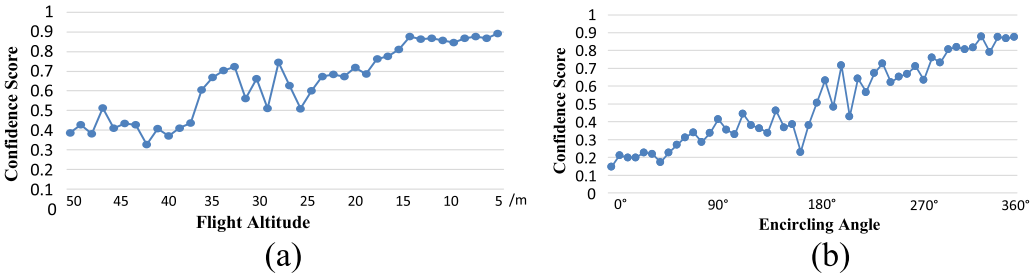


FIGURE B1 | Detection confidence for potential targets using oblique descending trajectory. (a) Real confidence. (b) Fitted confidence. [Color figure can be viewed at wileyonlinelibrary.com]

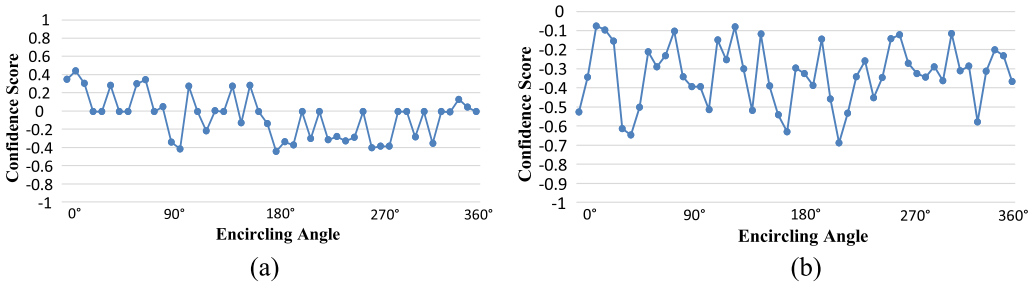


FIGURE C1 | Detection confidence of false target using circular trajectories at different altitudes. (a) 50 m altitude. (b) 5 m altitude. [Color figure can be viewed at wileyonlinelibrary.com]

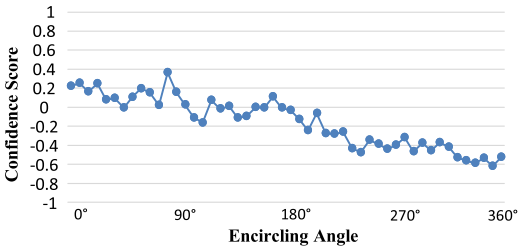


FIGURE C2 | Detection confidence of false target using oblique descending trajectory. [Color figure can be viewed at wileyonlinelibrary.com]

Appendix C

Confidence Fitting for False Objects

As shown in Figure C1a, the confidence distribution of false targets at 50 m altitude does not follow a single Gaussian distribution. Instead, it fluctuates around two distinct means, indicating a mixture of Gaussian distributions. Negative values represent confidence corresponding to other object classes. This is because false targets only partially resemble real rescue targets and may be misclassified from certain angles, while being correctly identified from others. Since this study mainly focuses on rescue targets. When fitting the false targets, only the part with positive confidence at a height of 50 m is considered, and a Gaussian distribution with a positive mean is used for fitting. Figure C1b shows that at 5 m altitude, target features are more obvious, and the overall confidence is negative, indicating a stronger tendency to classify as nonrescue targets. The fitting is performed using a Gaussian distribution without the Sigmoid function and with a negative mean. The variance settings refer to Appendix B.

The detection confidence for false targets using an oblique descending trajectory is shown in Figure C2. As the target distance decreases, the detection confidence generally declines and eventually stabilizes. The remaining characteristics and fitting methods are the same as those described in Appendix B.

Overall, the detection confidence for false targets is relatively low. Setting $\varepsilon > 0.6$ can effectively reduce the probability of false positives.

15564967, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/rob.70191 by Zhejiang University, Wiley Online Library on [02/06/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License